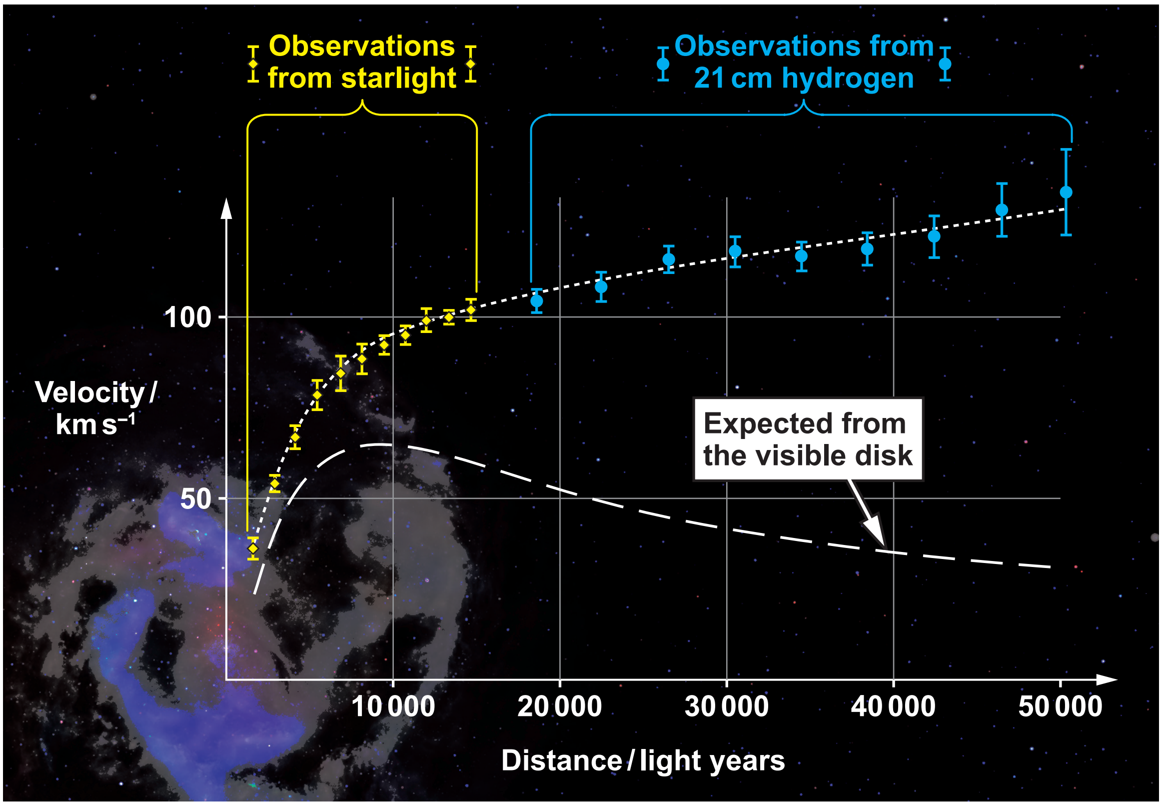
Examiner only

2. Light from a spiral galaxy is analysed and the following graph of velocity against distance from the centre of the galaxy is obtained.



(a) Show that the speed, v, of an object in a circular orbit of radius, r, about a massive object of mass, M, is given by: [3]

$$v = \sqrt{\frac{GM}{r}}$$

....

....

....

....

....

....

Examiner only

(b) Hence, explain why the graph is considered to be evidence for dark matter. [3]

(c) The lower curve in the graph suggests that an object a distance of 20 000 light years from the centre of the galaxy should have an orbital speed of approximately 50 km s^{-1}. Use this data to estimate the visible mass of the galaxy. (1 light year = 9.46×10^{15} m) [3]

(d) The observed data at 30 000 light years from the galactic centre were obtained using microwaves of wavelength 21 cm. Calculate the approximate wavelength shift that was observed to obtain the data plotted at 30 000 light years. [3]

(e) Use the Hubble equation to calculate the distance, from Earth, for a galaxy to have the same recessional speed as that of part (d). [2]

1420U401 05

14